

## --PREREQUISITE TASK

### **Roles and privileges**

Role is a single concept that covers both "users" and "groups" - the distinction is only whether the role has the LOGIN attribute. Roles are defined at the cluster level (stored in the shared catalog pg\_authid), so a role exists across every database in the instance, not inside one database.

A role has two kinds of rights:

Role attributes: LOGIN (can connect), SUPERUSER (bypasses all checks), CREATEDB, CREATEROLE, PASSWORD, etc.

Privileges on objects - what the role may do to a specific table, schema, sequence, or function, granted with GRANT and removed with REVOKE (GRANT SELECT, INSERT ON <table> TO <role>).

By default a role can only access objects it owns; to use objects in another schema it needs USAGE on the schema plus the relevant privilege on the object. Roles can also be members of other roles, which is how group-style permission management works. Because roles are cluster-level, creating one twice fails (role already exists) and dropping one requires that it no longer owns objects or holds grants.

A database is the top level for organizing objects; a client connects to exactly one database per connection and cannot directly reach objects in another (foreign data wrappers/dblink are needed for that). The first database, postgres, is created by initdb; template1 is cloned for every new CREATE DATABASE, so objects added to template1 propagate to all future databases.

### **Tablespace and database creation**

A tablespace is a named link to a directory in the filesystem where objects (tables, indexes) are physically stored. Two are created automatically:

pg\_default — the default location (under PGDATA/base); anything created without specifying a tablespace goes here.

pg\_global — holds the cluster-wide shared system catalogs.

A database is assigned a default tablespace, which can be changed with ALTER DATABASE ... SET TABLESPACE.

### **Database Cluster Data Directory Layout**

All objects are identified internally by OIDs (4-byte object identifiers), and the relationships between objects and OIDs live in the system catalogs (regular tables in pg\_catalog). A running

server manages a cluster - a set of databases sharing one configuration, one port, one filesystem location (PGDATA), and one set of background processes and shared memory.

The main directories under PGDATA:

base/ — one subdirectory per database, named by the database's OID; each contains the data files for that database's objects (a table maps to a file named by its OID, split into 1 GB segments when large).

global/ — the cluster-wide shared catalogs (roles, databases, tablespaces) that live in pg\_global.

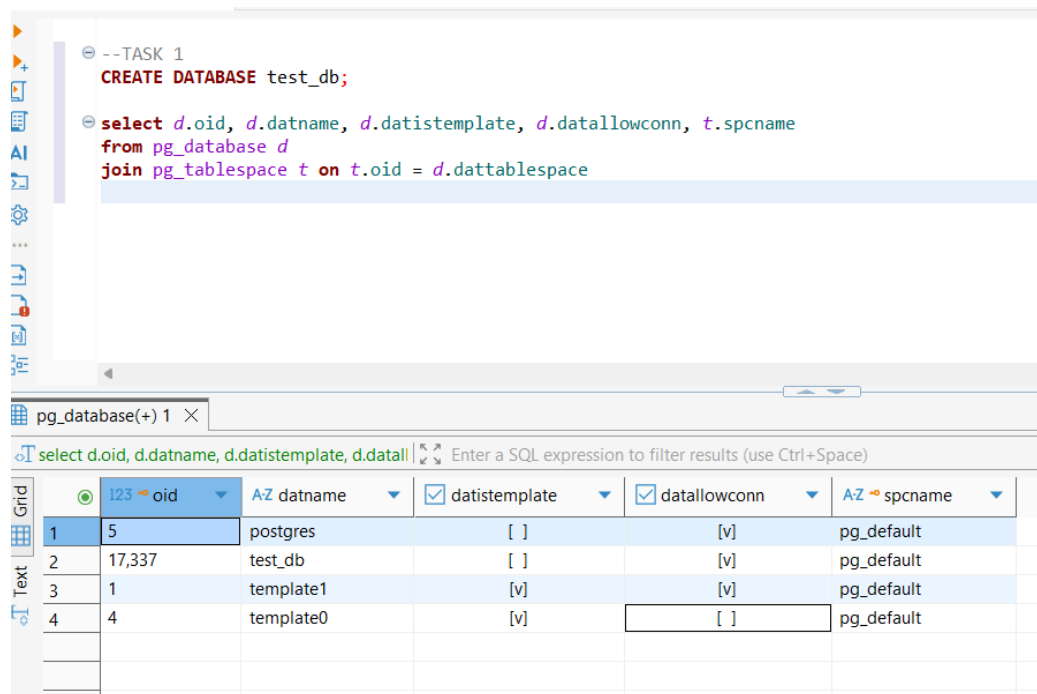
pg\_wal/ — the Write-Ahead Log segments (durability/recovery).

pg\_tblspc/ — symbolic links to any user-created tablespaces located outside PGDATA.

Configuration files: postgresql.conf, pg\_hba.conf, pg\_ident.conf.

## --TASK 1 CREATE NEW DATABASE

The test\_db was created to the default tablespace pg\_default.



The screenshot shows a PostgreSQL client interface with a SQL editor and a results grid. The SQL editor contains the following queries:

```
--TASK 1
CREATE DATABASE test_db;

select d.oid, d.datname, d.datistemplate, d.dataallowconn, t.spcname
from pg_database d
join pg_tablespace t on t.oid = d.dattablespace
```

The results grid displays the output of the second query, showing the OID, name, and tablespace for the databases in the cluster.

|   | oid    | datname   | datistemplate | dataallowconn | spcname    |
|---|--------|-----------|---------------|---------------|------------|
| 1 | 5      | postgres  | [ ]           | [v]           | pg_default |
| 2 | 17,337 | test_db   | [ ]           | [v]           | pg_default |
| 3 | 1      | template1 | [v]           | [v]           | pg_default |
| 4 | 4      | template0 | [v]           | [ ]           | pg_default |

## --TASK 2 CREATE NEW TABLESPACE

The new tablespace was created and test\_db was moved to it.

\*<postgres> Script-7 X

```

CREATE DATABASE test_db;

select d.oid, d.datname, d.datistemplate, d.dataallowconn, t.spcname
from pg_database d
join pg_tablespace t on t.oid = d.dattablespace

--TASK 2 CREATE NEW TABLESPACE

CREATE TABLESPACE mytablespace LOCATION 'C:\Program Files\PostgreSQL\18\data\tblspc_test\
ALTER DATABASE test_db SET TABLESPACE mytablespace

select * from pg_tablespace

```

pg\_database(+) 1 X

select d.oid, d.datname, d.datistemplate, d.dataallowconn, t.spcname Enter a SQL expression to filter results (use Ctrl+Space)

|   | 123 oid | AZ datname | <input checked="" type="checkbox"/> datistemplate | <input checked="" type="checkbox"/> dataallowconn | AZ spcname   |
|---|---------|------------|---------------------------------------------------|---------------------------------------------------|--------------|
| 1 | 5       | postgres   | [ ]                                               | [v]                                               | pg_default   |
| 2 | 1       | template1  | [v]                                               | [v]                                               | pg_default   |
| 3 | 4       | template0  | [v]                                               | [ ]                                               | pg_default   |
| 4 | 17,337  | test_db    | [ ]                                               | [v]                                               | mytablespace |

## --TASK 3 CREATE NEW SCHEMA

The schema 'lab' was created and the search route was directed to it.  
Initial data inserted.

\*<postgres> Script-7 X

```
--TASK 3 CREATE NEW SCHEMA

CREATE SCHEMA labs;

CREATE TABLE labs.person (
    id integer NOT NULL,
    name varchar(15)
);

SELECT schemaname, tablename FROM pg_tables
WHERE tablename = 'person';

INSERT INTO person VALUES(1, 'Bob');
INSERT INTO person VALUES(2, 'Alice');
INSERT INTO person VALUES(3, 'Robert');

SET search_path TO labs, public;
```

Statistics 1 X

| Name         | Value                                                                                                                    |
|--------------|--------------------------------------------------------------------------------------------------------------------------|
| Updated Rows | 3                                                                                                                        |
| Execute time | 0.003s                                                                                                                   |
| Start time   | Tue Jul 07 22:21:11 EEST 2026                                                                                            |
| Finish time  | Tue Jul 07 22:21:11 EEST 2026                                                                                            |
| Query        | INSERT INTO person VALUES(1, 'Bob');<br>INSERT INTO person VALUES(2, 'Alice');<br>INSERT INTO person VALUES(3, 'Robert') |

## -- TASK 4 INVESTIGATE MVCC

Separate transaction blocks to insert, update, and delete rows were performed  
The transaction flow and records positions were reflected on `t_xmin` and `t_xmax` values.

```
BEGIN; INSERT INTO labs.person VALUES (4, 'John'); COMMIT;
```

Its 1 X

CT txmin, txmax, tctid, tu Enter a SQL expression to filter results (use Ctrl+Space)

|   | txmin | txmax | tctid | row_data        |
|---|-------|-------|-------|-----------------|
| 1 | 3103  | 0     | (0,1) | \x01000000 [+2] |
| 2 | 3103  | 0     | (0,2) | \x02000000 [+2] |
| 3 | 3103  | 0     | (0,3) | \x03000000 [+2] |
| 4 | 3104  | 0     | (0,4) | \x04000000 [+2] |

```
BEGIN; UPDATE labs.person SET name='Alex' WHERE id=2; COMMIT;
```

s 1 X

T txmin, txmax, tctid, tu Enter a SQL expression to filter results (use Ctrl+Space)

|  | txmin | txmax | tctid | row_data        |
|--|-------|-------|-------|-----------------|
|  | 3103  | 0     | (0,1) | \x01000000 [+2] |
|  | 3103  | 3105  | (0,5) | \x02000000 [+2] |
|  | 3103  | 0     | (0,3) | \x03000000 [+2] |
|  | 3104  | 0     | (0,4) | \x04000000 [+2] |
|  | 3105  | 0     | (0,5) | \x02000000 [+2] |

FOR UPDATE;

**BEGIN; DELETE FROM person WHERE id = 3; COMMIT;**

| 1 X                                                                                   |        |        |                   |
|---------------------------------------------------------------------------------------|--------|--------|-------------------|
| t_xmin, t_xmax, t_ctid, tu  Enter a SQL expression to filter results (use Ctrl+Space) |        |        |                   |
| t_xmin                                                                                | t_xmax | t_ctid | row_data          |
| 3103                                                                                  | 0      | (0,1)  | ▶ \x01000000 [+2] |
| 3103                                                                                  | 3105   | (0,5)  | ▶ \x02000000 [+2] |
| 3103                                                                                  | 3106   | (0,3)  | ▶ \x03000000 [+2] |
| 3104                                                                                  | 0      | (0,4)  | ▶ \x04000000 [+2] |
| 3105                                                                                  | 0      | (0,5)  | ▶ \x02000000 [+2] |

```
BEGIN;INSERT INTO person VALUES(999, 'Test'); COMMIT;
```

1 X

t\_xmin, t\_xmax, t\_ctid, tu Enter a SQL expression to filter results (use Ctrl+Space)

| t_xmin | t_xmax | t_ctid | row_data        |
|--------|--------|--------|-----------------|
| 3103   | 0      | (0,1)  | \x01000000 [+2] |
| 3103   | 3105   | (0,5)  | \x02000000 [+2] |
| 3103   | 3106   | (0,3)  | \x03000000 [+2] |
| 3104   | 0      | (0,4)  | \x04000000 [+2] |
| 3105   | 0      | (0,5)  | \x02000000 [+2] |
| 3107   | 0      | (0,6)  | \xe7030000 [+2] |

```
BEGIN; DELETE FROM person WHERE id = 999; COMMIT;
```

1 X

t\_xmin, t\_xmax, t\_ctid, tu Enter a SQL expression to filter results (use Ctrl+Space)

| t_xmin | t_xmax | t_ctid | row_data        |
|--------|--------|--------|-----------------|
| 3103   | 0      | (0,1)  | \x01000000 [+2] |
| 3103   | 3105   | (0,5)  | \x02000000 [+2] |
| 3103   | 3106   | (0,3)  | \x03000000 [+2] |
| 3104   | 0      | (0,4)  | \x04000000 [+2] |
| 3105   | 0      | (0,5)  | \x02000000 [+2] |
| 3107   | 3108   | (0,6)  | \xe7030000 [+2] |

## --TASK 5 INVESTIGATE VACUUM

VACUUM labs.person command identified the 3 dead row versions and marked them reusable.

The 'Sarah' record landed on the 'new' row.

VACUUM FULL labs.person command built a completely empty data file and copied *only* the live rows ("Bob", "John", updated "Alex" and "Sarah") over to it. The old file was deleted.

```
VACUUM labs.person;
```

```
SELECT t_xmin, t_xmax, t_ctid,  
       tuple_data_split('labs.person'::regclass, t_data, t_infom  
FROM heap_page_items(get_raw_page('labs.person', 0));
```

1 X

t\_xmin, t\_xmax, t\_ctid, tu Enter a SQL expression to filter results (use Ctrl+Space)

| t_xmin | t_xmax | t_ctid | row_data          |
|--------|--------|--------|-------------------|
| 3103   | 0      | (0,1)  | ▶ \x01000000 [+2] |
| [NULL] | [NULL] | [NULL] | NULL              |
| [NULL] | [NULL] | [NULL] | NULL              |
| 3104   | 0      | (0,4)  | ▶ \x04000000 [+2] |
| 3105   | 0      | (0,5)  | ▶ \x02000000 [+2] |
|        |        |        |                   |
|        |        |        |                   |

```
INSERT INTO person VALUES (5, 'Sarah');
```

1 X

t\_xmin, t\_xmax, t\_ctid, tu Enter a SQL expression to filter results (use Ctrl+Space)

| t_xmin | t_xmax | t_ctid | row_data          |
|--------|--------|--------|-------------------|
| 3103   | 0      | (0,1)  | ▶ \x01000000 [+2] |
| 3109   | 0      | (0,2)  | ▶ \x05000000 [+2] |
| [NULL] | [NULL] | [NULL] | NULL              |
| 3104   | 0      | (0,4)  | ▶ \x04000000 [+2] |
| 3105   | 0      | (0,5)  | ▶ \x02000000 [+2] |
|        |        |        |                   |
|        |        |        |                   |



**VACUUM FULL** labs.person;

| t_xmin, t_xmax, t_ctid, tuq   Enter a SQL expression to filter results (use Ctrl+Space) |        |        |                   |
|-----------------------------------------------------------------------------------------|--------|--------|-------------------|
| t_xmin                                                                                  | t_xmax | t_ctid | row_data          |
| 3103                                                                                    | 0      | (0,1)  | ▶ \x01000000 [+2] |
| 3109                                                                                    | 0      | (0,2)  | ▶ \x05000000 [+2] |
| 3104                                                                                    | 0      | (0,3)  | ▶ \x04000000 [+2] |
| 3105                                                                                    | 0      | (0,4)  | ▶ \x02000000 [+2] |